Module 2 Lab 2: Metric Selection Under Cost Pressure

Submission template. 20 points. Fill this document in as you work, then upload it to Canvas with your notebook and a PDF of the full notebook run. Check Canvas for the due date.

Your Unique Values

Your name	Ehmar Reza
Scenario you chose for Step 1 (scenario_id)	S3
GitHub portfolio repository URL	https://github.com/donkobonko/aws-ai-portfolio
Date and time you last ran your notebook top to bottom	11/1

Prediction 1: Accuracy Versus Recall

Write this BEFORE Step 1. Look only at the four counts for your chosen scenario. Will accuracy be higher or lower than recall, and by roughly how much? An honest wrong prediction earns full credit.

My prediction and reasoning	Prediction: well, if we were to "recall" the equations, accuracy has about three more terms in the equation. so that means whether or not recall has a higher or lower value than accuracy is mainly dependent on false negatives and true negatives, and if we look at the false negatives for S1 (my chosen scenario), then we'll see that there's only two false negatives, but there is a really high number of true negatives, meaning TN > FN, and TP < TN (making TP insignificant), so accuracy is likely to be higher in this scenario.

Step 1: One Scenario by Hand

No code, and cover the computed columns in the file. Show every division.

TP, FP, FN, TN for my scenario	S1 Hospital s 40 8 2 950
Accuracy, with arithmetic	(40 + 950) / (40 + 8 + 2 + 950) = 990 / 1000 = 0.990
Precision, with arithmetic	40 / (40 + 8) = 40 / 48 = 0.8333
Recall, with arithmetic	40 / 40 + 2 = 40 / 42 = 0.9524
F1, with arithmetic	2(0.833 * 0.953) / (0.833 + 0.953) = 0.8889

Do these match the file's columns within rounding? If not, what was the error?	Yeah they match
Prediction 1 compared with what I found	I actually made a really good guess and I'm kinda proud of myself for that one, not sure if my reasoning was actually strong but it worked well.

Step 2: All Four in Code

The code lives in your notebook. Record the result here.

Do my computed columns match the file for all four scenarios?	Yes.
Any mismatch I found and how I fixed it	Nope. (technically the mismatch would be the rounding, I'd just have to round to the ones in the CSV).

Prediction 2: Accuracy Versus Cost

Write this BEFORE you run Step 3. Will the scenario with the highest accuracy also have the lowest total error cost?

My prediction and reasoning	Prediction: Since all four terms in the equation are equally important, I'd say it's a little difficult to predict. The highest accuracy goes to S3, a total of 99.3%. However, S1 has some insane costs for false negatives, so they might trump the total error cost in general for any of them, and they have the second lowest accuracy. Now if we compare these guys to the the one with the lowest accuracy, S4, they seem to have a moderately higher combined cost per fp and cost per fn, meaning the final evaluated sum may be high, therefore I would say, yes, I think higher accuracy yields lower total error cost.

Step 3: Total Error Cost

Scenario	Accuracy	Total error cost $	Rank by cost (1 = highest)
S1	0.9900	500008	1
S2	0.9918	7520	4
S3	0.9930	27140	3

Scenario	Accuracy	Total error cost $	Rank by cost (1 = highest)
S4	0.9536	343000	2

Prediction 2 compared with what the table showed	Yeah okay so I'm a little dumb and didn't notice how close S1 and S2 were in accuracy and overlooked how large it is given that tiny difference. Also, S2 and S3 disproved it very clearly.

Step 4: Metric Call for Each Scenario

Write all four calls before you open the instructor key column.

Scenario	Metric I would optimize	Error type that drove the choice, in two sentences
S1	Recall	The cost for missing an actual case is extremely expensive. This would be damaging for the business and would thus need to optimize to catch significantly more negative cases.
S2	Precision	The cost for missing false positives is a little more expensive. Optimizing for precision here would yield better results so the model catches less false alarms.
S3	Recall	The cost for false negatives is somewhat high. Would need to optimize to catch more negative cases
S4	Recall	The cost for false negatives is considerable expensive. Need to optimize to catch more negatives and reduce the false counts.

After opening the key: where I disagreed, and why	Never ended up finding the key, so can't really do this. I'm assuming this was supposed to be the updated template… but nothing on canvas was updated.

Step 5: S4 Threshold Sweep

Model at least five settings. Each step adds 4 false positives and removes 1 false negative.

Step	False positives	False negatives	Total error cost $
0	130	95	343000

Step	False positives	False negatives	Total error cost $
1	134	94	341000
2	138	93	339000
3	142	92	337000
4	146	91	335000
5	150	90	333000

Step with the lowest total cost	5
Dollar saving against step 0	$10000

Reflection

In your own words, no AI assistance. Answer from your own run. Graders check these answers against your notebook outputs.

1. Report your Step 3 prediction and what the table actually showed. Quote your own accuracy and total error cost figures for at least two scenarios.	Okay well I didn't think I needed to calculate my own total error costs (maybe this question is just worded wrong or I read it wrong), but anyways, S2 and S3 prove the statement wrong really quick. Very clear difference in accuracy – S3 is higher buy a whole .12% difference, however S3 doesn't have a lower total error cost.
2. Scenario S1 has an accuracy of 0.99 in the file. Explain, using the actual counts, why that number is close to meaningless and what you would report to a hospital administrator instead.	Well honestly the best thing I can say is: risking a single false negative is devastating to expenses, as reporting false positives costs as much as there are false positives. I'd say that 99% accuracy is meaningless if costs are still taking severe hits.
3. Your S4 threshold sweep found a minimum. State the setting and the dollar saving against the starting point. Then say what would have to be true operationally for that setting to be a bad idea anyway.	The setting was 150 FP, 90 FN, and it saved $10000 against the starting point. Operationally, what that does is suddenly the machine is working with different data and has flagged a bunch of inaccurate stuff to account for desired needs, but that kind of produces bad results anyway.

AI Use Disclosure

List any AI tool you used, what you used it for, and how you checked what it produced. Write None if you used none. Predictions and reflection answers must be your own work with no AI assistance.

Tools used	Gemini
What I used them for	the vectorized Pandas column formulas for classification metrics, the threshold sweep arithmetic ($2,000 net savings per step from 4 FPs vs. 1 FN), and the verification that the instructor key was absent from the dataset.
How I verified the output	Cross referenced across lecture material, prevented it from auto-generating code, and used advice on using the dataframe to verify instructor key was in fact missing.

Before You Submit

- ☐ I computed one scenario entirely by hand before writing code
- ☐ My reflection quotes my own numbers, not the example numbers
- ☐ Both predictions were written before I ran the matching step
- ☐ I recorded the actual result for both predictions, including any I got wrong
- ☐ Notebook runs top to bottom with every output visible
- ☐ PDF exported after that full run and it matches
- ☐ Work pushed to my GitHub portfolio repository and the URL is above
- ☐ Reflection answers written by me with no AI assistance
- ☐ Files uploaded individually, no archive